



## **ANNEX 1 APPLIED IN DRUG SUBSTANCE FACILITIES**

**A risk-based approach to  
design, operations and  
grades in low bioburden  
drug substance (DS)  
manufacturing—industry  
interpretation of Annex 1**



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Since its inception in 2004, BioPhorum has become the open and trusted environment where senior leaders of the biopharmaceutical industry come together to openly share and discuss the emerging trends and challenges facing their industry.

Growing from an end-user group in 2008, BioPhorum's membership now comprises top leaders and subject matter experts from global biopharmaceutical manufacturers and suppliers, working in both long-established and new Phorums. They articulate the industry's technology roadmap, define the supply partner practices of the future, and develop and adopt best practices in drug substance, fill finish, process development and manufacturing IT.

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## Executive summary

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**This paper offers guidance to drug substance (DS) manufacturers on interpreting and applying Annex 1 of the EU GMP guidelines, typically for sterile products, to their low bioburden manufacturing processes which are primarily governed by Annex 2. It addresses the design of premises, operational grades and contamination control strategies (CCS), advocating for a risk-based approach rather than a strict application of Annex 1's sterile manufacturing requirements.**

The paper discusses cleanroom classifications, barrier technologies, closed systems and controlled non-classified (CNC) spaces, aiming to provide clarity and promote consistency within the low bioburden DS manufacturing industry while ensuring patient safety and product quality without unnecessary over-application of stringent sterile guidelines.

## Introduction

Biological (low bioburden) drug substance (DS) manufacturing facilities that are based in or supply into Europe are governed by Annex 2, Manufacture of Biological Active Substances and Medicinal Products for Human Use<sup>1</sup> and the general guidelines in EudraLex—Volume—Good Manufacturing Practice (GMP)<sup>2</sup>. However, the current (recently revised) version of Annex 1, Manufacture of Sterile Medicinal Products<sup>3</sup>, mentions that guidance pertaining to sterile manufacturing operations and products could also be applied to low bioburden manufacturing facilities.

This version of Annex 1, published in August 2022, states:

*“The intent of the Annex is to provide guidance of the manufacture of sterile products. However, some of the principles and guidance, such as contamination control strategy, design of premises, cleanroom classification, qualification, validation, monitoring and personnel gowning, may be used to support the manufacture of other products that are not intended to be sterile such as certain liquids, creams, ointments, and low bioburden biological intermediates, but where the control and reduction of microbial, particulate and endotoxin/pyrogen contamination is considered important”.*

Given that both Annexes 1 and 2 are adopted by Pharmaceutical Inspection Convention and Pharmaceutical Inspection Co-operation Scheme (PIC/S), and the same guidance would be relevant for all PIC/S member countries beyond the European countries under EudraLex, it was the opinion of the low bioburden manufacturing industry who are

members of BioPhorum that additional clarity should be provided as to the relevancy and applicability of Annex 1 to low bioburden DS manufacturers held to Annex 2 requirements. Alignment across low bioburden DS manufacturers on this topic is essential to ensure consistency and provide everyone, including regulators, with a risk-based roadmap.

An overarching theme in the current version of Annex 1 is the requirement to document and implement a comprehensive CCS. Design of Premises/Operations and Grades is a component of the CCS; in this paper we share an interpretation of how Annex 1 requirements for Design of Premises/Operations and Grades can be applied to the low bioburden manufacturing industry, this is especially important as Annex 2 refers to quality risk management (QRM) principles and Annex 1 for the selection of environmental classification cascades and associated controls (Premises and Equipment, point 6) and establishment of a control strategy to protect the product (Starting and Raw Materials, point 33).

## 2.0

# Purpose and scope

**This paper was written collaboratively by a team of experts from 11 companies in the biopharmaceutical industry, facilitated by BioPhorum, who have a history of assessing potential for new technologies to benefit industry. This paper is a consensus response to Annex 1, representative of current thinking across the industry; it does not represent the internal policies, views or opinions of the authors' respective individual companies.**

The purpose of this paper is twofold: to share an interpretation of how the requirements of Annex 1 for Design of Premises/Operations and Grades can be applied to low bioburden DS manufacturing facilities and processes, and to provide enhanced clarity for the industry. This paper supports a scientific risk-based approach that identifies low bioburden DS manufacturing industry requirements, and the types of control that meet those requirements. While this paper contains a consensus of opinions generated by a sample of low bioburden manufacturing

representatives, it is not exhaustive of every manufacturing process and conditions.

It should be noted that low bioburden DS manufacturing facilities can utilize different classification terminologies (e.g. International Organization for Standardization (ISO) classifications)<sup>8</sup>. In this paper, 'grades A to D' is the terminology used. The low bioburden manufacturing facility may have a document detailing the comparability of their respective classification terminologies, e.g. ISO Classification to EU Grading (grades A to D).

# 3.0

## Design of premises for DS manufacturing

The current version of Annex 1 puts greater emphasis on CCS. The updated Annex 1 also puts greater emphasis on QRM principles aligned to ICH Q9, which are already present in Annex 2. Accordingly, risk assessment of the facility design, equipment and processes should be undertaken to identify potential hazards that could impact product quality. There is a clear link between product, process and facility, with process and risk assessment driving facility design. A key factor in contamination control is design of DS manufacturing facilities.

Annex 1 includes additional information and clarification in Section 4. This is summarized below with interpretations of potential application in low bioburden DS manufacturing.

### Airlocks

Given that cleanrooms provide protection for the product, transitions in and out of cleanrooms are highly important to maintaining the integrity of the intended cleanroom environment. As stated in 4.10:

*"The transfer of equipment and materials into and out of the cleanrooms and critical zones is one of the greatest potential sources of contamination. Any activities with the potential to compromise the cleanliness of the cleanrooms or the critical zone should be assessed and if they cannot be eliminated, appropriate controls should be implemented".*

Section 4 Premises provides detailed information as to the inclusion of passthroughs and airlocks for personnel materials, equipment.

These include:

- Unidirectional flows for grade A or B areas (4.11)
- Segregation of material and personnel airlocks wherever possible (4.12)
- Cleaning and disinfection commensurate with the risk and in line with the CCS (4.12 ii)
- And describes gowning in "areas of increasing cleanliness used for entry of personnel" (4.12 i).

While the guidance is specific to EU grade and for sterile products, low bioburden premises also require airlocks, where applicable, to protect the manufacturing environment and non-sterile products, which coincide with the general principle of protecting the cleanroom environment from ingress of contaminants and guidance for grades C and D.

Where EU grade applies, the airlocks to these grade D, C areas are typically provided in accordance with the guideline, with filtered air of an appropriate cleanliness and application of airlocks from hallways to grades.

Facilities classified via ISO standard also apply airlocks and passthroughs to protect the manufacturing area but adapt room air pressurization cascades through one airlock to facilitate more direct access to a grade C-equivalent area from a CNC hallway. The area classification for the airlock meets at rest, the grade applicable for the manufacturing area it serves.

Grade A and B and material sterilization, however, are not very common for bioburden-controlled DS manufacturing and therefore exclusively unidirectional flows are less common.

- Personnel airlocks are segregated from material entry airlocks, wherever feasible
- Waste flow is unidirectional via physically separated exit airlock, or a procedural control is in place for bidirectional material airlock (MAL) to protect the incoming materials
- Bidirectional flow personnel within an entry/exit personnel airlock (PAL) is typical, especially for low traffic areas, where procedural controls are in place to prevent simultaneous gowning/de-gowning
- Passthroughs for material flow are generally passive in design, if supporting low risk material or closed processes.



The transfer of materials from uncontrolled to cleanroom environments represents a key contamination risk factor, and therefore MALs are typically designed to ensure a microbial reduction step (e.g. wipe down with a sanitization solution) to mitigate the risk for transfer to the more critical areas.

- The stringency specified for inventory management of approved lists of materials subject to material transfer to the cleanroom (per Section 4.12 ii) is generally not applicable, as long as the materials being transferred are adequately cleaned/sanitized for entry
- Where high-risk steps apply (e.g. open operations), the material sanitization requirements for materials increase accordingly.

### 3.1 Barrier technologies

In Annex 1 facilities, use of isolators and restricted access barrier systems (RABS) for protection of critical zones is the baseline expectation for facilities producing sterile products. Section 4.3 states: “any alternative approaches to the use of RABS or isolators should be justified”.

However, when it comes to high-risk operations for DS manufacturing, like open inoculum preparation requiring low bioburden control, applying the full extent of Annex 1’s stringent guidance around designing, qualifying, maintaining and monitoring grade A environments suitable for aseptic processing is likely an overapplication. It is recommended that an appropriate level of local protected environment (LPE) is provided, based on risk assessment and CCS (this is discussed more in the later section of this paper on LPE).

It is worth noting that barrier technologies, such as isolators and RABS, are clean air equipment, comparable to cleanrooms. Barrier technologies are a different category of equipment to closed systems and single-use systems (SUS), which are included in Section 8 (*Production and specific technologies*).

### 3.2 Cleanroom and clean air equipment qualification

Section 4 (*Premises*) includes a new subsection, *Cleanroom and clean air equipment qualification*. It establishes requirements for initial qualification of cleanroom and clean air equipment, including pressurization, airflow velocity, smoke study and requalification frequency.

Generally, higher grade and LPEs require more qualification activities, more frequent requalification and more detailed documentation. Therefore, it is important to select the appropriate classification for each operation in the DS facility and understand identified risks. This approach will ensure unnecessary engineering controls, operational burdens and capital expenses are reduced. LPEs requiring unidirectional airflow (e.g. biological safety cabinets (BSCs)) can benefit from dynamic smoke pattern testing to confirm equipment functionality under routine conditions, especially to demonstrate the protecting of the clean zone from lower background environments, similarly passthroughs and airlocks can also sometimes benefit from dynamic smoke pattern testing.

Related, but different from Section 4, Section 9 *Environmental and process monitoring* includes guidance for ongoing routine monitoring, setting action limits, alert levels and reviewing trend data. BioPhorum has published a similar white paper entitled *Streamlining environmental monitoring in low bioburden drug substance production: industry interpretation of Annex 1<sup>4</sup>*.

### 3.3 Risk-based approach

Aligned with Annex 1's overarching QRM approach, a publication in the PDA Journal, *Risk-Based Selection of Environmental Classifications for Biopharmaceutical Operations*<sup>5</sup>, proposes a scientific risk-based approach when assigning grades to cleanrooms used at low bioburden manufacturing sites.

The model focuses on the likelihood of a microbial ingress and proliferation outside process limits based on:

- Bioburden limit
- Level of closure of the system
- Growth potential in terms of time, nutrients and temperature
- Ability of the process to detect contamination.

Even though the risk-based approach provides a detailed evaluation model of the potential microbial proliferation, the risk of non-viable particulates and pyrogen, and adventitious agents (bacteria, fungi virus, mycoplasma etc.) should also be considered in the CCS. Annex 1 Section 8 includes a description of closed systems and their value in increasing product protection. Similarly, low bioburden processes are increasingly adopting closed processes based on technological advances. While a closed system provides a robust control to prevent any kind of contamination related to the surrounding environment, the non-viable particulate contamination controls for equipment that is closed via clean-in-place/steam-in-place (CIP/SIP) and/or open operations in controlled conditions must be evaluated. The manufacturer should provide a holistic process view on process controls to minimize the possibility of particle contamination (e.g. clean steam condensation during SIP, vessel rinse before SIP, equipment cleaning, product filtration).

This holistic approach aligns with GMP principles in Annex 2, which states:

*"The application of appropriate environmental controls and monitoring and, wherever feasible, in-situ cleaning and sterilization systems together with the use of closed systems can significantly reduce the risk of accidental contamination and cross-contamination".*

Other approaches for design of low bioburden DS manufacturing facilities can also be considered:

- The first step in designing the facility layout is often manufacturing process flow mapping. This includes identifying the sequence of operations, equipment and the required environmental controls for each step.
- Equipment used in the manufacturing process should be selected and designed to minimize the risk of contamination. This could include the use of closed systems, aseptic connectors and single-use equipment. The placement of equipment and processes should also be considered.
- The flow of materials and personnel should be designed to minimize the risk of contamination. This includes understanding the entry and exit points for personnel, as well as transfer areas for raw materials, in-process materials, finished products and waste materials.
- The facility design and flows should minimize the risk of cross-contamination and mix-up where multiple batches or multiple products are produced concurrently.
- The facility layout should be designed to facilitate cleaning and maintenance activities. This includes designing facility surfaces that are easy to clean and disinfect, and providing adequate space for equipment maintenance.

For existing facilities, where not all design elements in the latest version of Annex 1 may have been considered, a risk assessment should be undertaken, including a detailed review of historical data and documentation to identify potential hazards, risks and vulnerabilities in the manufacturing environment. This assessment will help determine whether improvements are required and guide the decision-making process.

# 4.0

## Grades and operations for DS manufacturing

**Grades and operations are interdependent, working together to prevent contamination of products. Annex 1 and Annex 2 are both aligned on the principle that environmental classification must respond appropriately to the operation.**

Annex 2, point 6 acknowledges the importance of adapting environmental control to the specific product and process, emphasizing use of QRM principles to determine the appropriate level of control. It does not provide specific cleanroom classifications or monitoring limits; for these, it refers to Annex 1:

*"Where processes are not closed and there is therefore exposure of the product to the immediate room environment [...] control measures should be put in place, including engineering and environmental controls on the basis of QRM principles. These QRM principles should take into account the principles and guidance from the appropriate sections of Annex 1 to EudraLex, Volume 4, when selecting environmental classification cascades and associated controls".*

This point is followed by footnote 13 that clarifies: *"it is not the intention to force the manufacture of sterile product at a stage when a low bioburden is appropriate and authorised. Its use is because it is the only EU GMP source of guidance on all of the classified manufacturing areas including the lower grades D and C."*

Annex 1, point 4.23 states:

*"Each manufacturing operation requires an appropriate environmental cleanliness level in the operational state in order to minimize the risk of contamination of the product or materials being handled".*

Principles and guidance on environmental classification and associated controls are scattered throughout Annex 1, including in sections 4 (*Premises*), 5 (*Equipment*), 7 (*Personnel*), 8 (*Production and specific technologies*) and 9

(*Environmental and process monitoring*). However, referring to Annex 2, point 6 and the clarification in footnote 13, this paper advises focusing more on principles of contamination control through environmental classification and less on specific guidance written for manufacture of sterile products.

Annex 1, Section 4 (*Premises*) introduces environmental classification. Point 4.4 lists the four grades (grades A to D) of cleanroom/zone for the manufacture of sterile products. The *Cleanroom and clean air equipment qualification* section includes Table 1 and Table 2, which establish the maximum particle concentration and microbial contamination level for each grade, respectively. These tables are intended for cleanroom qualification, and corresponding tables for monitoring and action limits are found in Section 9 (Table 5 and Table 6).

Section 8 (*Production and specific technologies*) provides guidance on how these grades correspond to various examples of operations. Annex 1 provides two separate tables for illustrating examples of operations and grades, Table 3 for terminally sterilized products (lower risk profile), and Table 4 for aseptic preparation and processing (higher risk profile).

This principle of aligning environmental classification with the risk profile of specific operations should be adopted for bioburden-controlled DS preparation and processing operations. The degree of environmental control of particulate and microbial contamination of the production premises can be adapted to the product type, production step and process closure. Ultimately, the manufacturer's CCS will document the controls needed to maintain product quality and patient safety in a cost-efficient manner.

The members of BioPhorum did try to produce a table linking operations in low bioburden biologic DS manufacturing to the appropriate cleanroom grades. However, this became very complex due to the variation seen within member organizations. It is advised that sites produce a table linking the operation to the risk and then the appropriate cleanroom classification, i.e. operation of a closed bioreactor = low risk = non-classified space, open bulk filling = medium risk = grade C, open inoculum operations = high risk = LPE.

Selection of environmental classification based on risk assessment should include both core process and support operations as stated in Annex 2, point 33:

*"Given that risks from the introduction of contamination and the consequences to the finished product is the same irrespective of the stage of manufacture, establishment of a control strategy to protect the product and the preparation of solutions, buffers and other additions should be based on the principles and guidance contained in the appropriate sections of Annex 1".*

The following sections explore systems and environmental conditions commonly utilized for low bioburden DS manufacturing.

## 4.1 Local protected environment (LPE)

Where operations are subjected to a significant contamination risk when exposed to the environment, LPE systems may be used to create a clean-air protected area immediately surrounding the operations. General principles of first air and barrier technologies from Annex 1 for the protection of aseptic environment through separation from the surrounding room can be relevant for LPE. However, applicable sections of Annex 1 guidance (if any) should be carefully selected based on identified risk to the low bioburden process.

System type, air quality within the LPE and associated controls (e.g. gowning requirements and materials transfer in/out of the LPE zone) should be selected based on the scale and ergonomics of the operations and risk of contamination to the product. If the risk of contamination

is determined to be very high (e.g. aseptic handling such as open inoculum operations), air quality is expected to be high and a critical LPE (CLPE) should be used. CLPE are designed to meet grade A conditions at rest. However, the non-viable and viable limits in operation must be defined based on QRM.

LPE can vary in its level of barrier separation, from limited separation like walk-in downflow booths where personnel occupy the LPE space and the LPE zone is open to the background environment, to full separation like a closed isolator where the LPE enclosure is fully sealed and isolated from the external environment (background cleanroom air and personnel). Examples of LPE include BSCs, unidirectional downflow hoods (UDFHs), fume hoods, RABS and isolators.

Annex 1 states the minimum background environment classifications required for aseptic processing operations within BSC (grade B), RABS (grade B), open isolator (grade C) and closed isolator (grade D). However, for low bioburden operations, the background environment classification should be selected based on QRM principles rather than direct application of Annex 1 requirements. What is important is ensuring the background environment does not adversely impact the critical activity within the LPE or CLPE.

While the main purpose of the LPE is to provide a clean protected area around high contamination risk operations, it offers the additional benefit of reduced capital and operating costs. With the LPE of high classification limited to a small area immediately surrounding the high-risk operation, the rest of the cleanroom can be classified at a lower grade, resulting in construction of lower-grade cleanroom finishes, lower volume of high-classification air, less gowning, material transition, cleaning and monitoring requirements. For more details on environmental monitoring refer to the BioPhorum paper *Streamlining environmental monitoring in low bioburden drug substance production: industry interpretation of Annex 1*<sup>4</sup>.

## 4.2 Closed systems

Definitions of closed systems are:

Annex 1: "A system in which the product is not exposed to the surrounding environment [...] Closed systems are not opened until the conclusion of an operation". The use of the term 'closed systems' in Annex 1 does not refer to systems such as RABS or isolator systems.

Annex 2: Where a drug substance or product is not exposed to the immediate room environment during manufacture.

ISPE Baseline Guide<sup>6</sup>: A system that isolates the process zone from its surrounding environment and prevents ingress of environmental contaminants during product contact. The process zone is limited by equipment. System closure requires two steps:

1. Removal of latent contaminants within the process zone to a level commensurate with the process requirements, whether sterile, aseptic, or low bioburden, prior to process contact.
2. Prevention of ingress of environmental contaminants during process contact.

Note: A closed system is not necessarily a sterile system. It is closed to the appropriate condition whether sterile, aseptic or low bioburden.

### Annex 1 and interpretation for low bioburden DS

Annex 1 includes specific guidance for closed systems in Section 8 (*Production and specific technologies*). Each of the four Annex 1 paragraphs under *Closed systems* (8.127–8.130) and interpretation for application to low bioburden DS are as follows:

#### Paragraph 8.127

*The use of closed systems can reduce the risk of microbial, particle and chemical contamination from the adjacent environment. Closed systems should always be designed to reduce the need for manual manipulations and the associated risks.*

#### Interpretation for application to low bioburden DS

The use of closed systems in DS manufacture can significantly enhance an organization's CCS. These systems should be designed to minimize manual manipulation for reducing the risk of connection errors or defects leading to a system integrity breach.

#### Paragraph 8.128

*It is critical to ensure the sterility of all product contact surfaces of closed systems used for aseptic processing. The design and selection of any closed system used for aseptic processing should ensure maintenance of sterility. Connection of sterile equipment (e.g. tubing/ pipework) to the sterilised product pathway after the final sterilising grade filter should be designed to be connected aseptically (e.g. by intrinsic sterile connection devices).*

#### Interpretation for application to low bioburden DS

This specifically references the requirements for aseptic processing. With respect to DS manufacture, product contact surfaces and connection types will be selected and assessed to assure the process meets the product safety requirements and are commensurate with process requirements, whether sterile, aseptic or low bioburden, prior to process contact. Annex 1 Table 4 describes an example of closed technology requiring more stringent background classification for "assembly of closed and sterilised SUS using intrinsic sterile connection devices" being required in grade D. However, the interpretation is directly applicable to sterile medicinal products rather than low bioburden processes.

#### Paragraph 8.129

*Appropriate measures should be in place to ensure the integrity of components used in aseptic connections. The means by which this is achieved should be determined and captured in the CCS. Appropriate system integrity tests should be considered when there is a risk of compromising product sterility. Supplier assessment should include the collation of data in relation to potential failure modes that may lead to a loss of system sterility.*

#### Interpretation for application to low bioburden DS

Assurance of connection integrity will be a consideration in the design of a closed system to minimize risk of an integrity breach. Integrity assurance controls and failure data (and numerous other factors) influence the probability of failure scoring in a risk-based closure assessment.

### Paragraph 8.130

*The background environment in which closed systems are located should be based on their design and the processes undertaken. For aseptic processing and where there are any risks that system integrity may be compromised, the system should be located in grade A. If the system can be shown to remain integral at every usage (e.g. via pressure testing and/or monitoring) then a lower classified area may be used. Any transfer between classified areas should be thoroughly assessed (see paragraph 4.10). If the closed system is opened (e.g. for maintenance of a bulk manufacturing line) then this should be performed in a classified area appropriate to the materials (e.g. grade C for terminal sterilization processes, or grade A for aseptic processing) or be subject to further cleaning and disinfection (and sterilization in case of aseptic processes).*

### Interpretation for application to low bioburden DS

The background environment for closed system manufacturing will be considered in the risk-based closure assessment for DS manufacture.

Identification and assessment of planned and non-planned (failure mode) interventions will form part of the process closure risk assessment with defined mitigation implemented to minimize any risk to product safety. Consideration of environmental controls for execution of identified interventions will form part of the assessment, as will pre-use conditioning of process contact surfaces to ensure contamination control of the process.

### Risks and benefits

Annex 2 states:

*The application of appropriate environmental controls and monitoring and, wherever feasible, in-situ cleaning and sterilization systems together with the use of closed systems can significantly reduce the risk of accidental contamination and cross-contamination.*

Closed systems are already widely used in DS manufacturing facilities and are key considerations in facility design and cleanroom grade selection. For open processing, a highly protected environment is needed to reduce the risk of contaminating the product. But for closed processes, it makes sense to operate in facilities with less strict requirements as the environment does not pose a risk to the process. Closed systems significantly reduce the risk of contamination to the process stream, which means the environmental classification becomes less significant in the overall CCS.

Implementation of closed processing and the associated reduction in dependency on the environmental classification for operation offers numerous potential benefits:

- Prevention of ingress of contaminants to the process stream, reduction in overall contamination risk profile, increased patient safety and security of supply for life-changing medicines
- Wider environmental benefits associated with reduced energy usage achieved from reduced requirements to operate air-handling units to maintain the classified environment
- Reduced environmental monitoring effort associated with performance assessment of classified environment (driven by risk assessment output)
- Fewer airlocks for space savings and operational efficiency
- Reduced gowning for production personnel.



Theoretically, if the closure of a process is validated, the risk of ingress of external contaminations can be excluded, regardless of the classification of the immediate environment. There are various risks to take into consideration when operating a closed process and understanding these risks is important for the selection of the environment.

A risk assessment should be performed to understand the impact and implication in terms of the product and the patient. A closure risk assessment should be performed, and maintenance of the risk assessment should be an iterative process. The more connections and inlets/outlets in the process, the more complicated the risk assessment, as they add to the number of potential breach/failure points that need to be assessed.

There are also risks related to production staff. The mindset of people working in the facility may change with less strict room requirements. Production staff may begin to believe that they can do anything without compromising the product or production process, thereby increasing the risk of breaches or contamination events. Furthermore, if staff work in several facilities, it may be difficult for them to keep the right mindset for each of the facilities, if these are not operated in a similar manner.

### 4.3 CNC

Controlled-non/not-classified/clean-not-classified (CNC) space is only appropriate for closed processes with minimal risk of contamination from the surrounding environment as it is unclassified. This is discussed in greater detail in ISPE Baseline Guide<sup>6</sup> and BioPhorum's *Closed systems in CNC space—Closure playbook*<sup>7</sup>. There is no mention of either CNC or uncontrolled areas in the latest version of Annex 1, even though there were some definitions in earlier draft versions of the guidance and these terms are used widely in the industry.

CNC can be broadly described as an unclassified environment where user-defined parameters are monitored and controlled to user-defined limits. Based on International Society for Pharmaceutical Engineering (ISPE) and other guidelines, common controls include use of cleanable industrial finish with floor designed to contain spills, procedural controls, good housekeeping practices, access controls, personnel garments and filtered ventilation air, and ability to meet grade D requirements. CNC heating, ventilation and air-conditioning systems (HVAC) systems are usually designed to reduce airborne contaminants (viable, non-viable) below that of a non-GMP (e.g. administrative) environment. Both temperature and relative humidity (RH) can be controlled more tightly than in the ambient environment depending on the product and process requirements. Claims for environmental control in these areas are related to the design of the system and installation qualification is common. However, there is often no claim made or qualification executed in relation to non-viable particulate control. Typical systems will have user-defined heating, cooling and filtration of supply air meeting a minimum efficiency reporting value (MERV) (e.g. using an appropriate user-defined MERV classification).

As levels of finish, control and facility cleaning should correspond to required levels of process/product protection and manufacturer practices, it is difficult to establish specific requirements and controls for CNC areas. Ultimately, the manufacturer's CCS should list the architectural finishes and controls required to provide the protection level needed to preserve the product quality for the processes being carried out.

Closed processing environment (CPE) is another term used to describe a manufacturing environment that houses closed systems and processes. Like CNC, appropriate environmental controls and specifications are user defined and commensurate with risk to process and product.

## Conclusion

**While the stated intent of Annex 1 is to provide guidance for the manufacture of sterile products, some of the key principles and guidance might be applicable to low bioburden DS manufacturing processes. As such, the aim of this paper is to provide the low bioburden DS manufacturing industry with additional guidance and discussion surrounding which 'Design of Premises/Operations and Grades' principles from Annex 1 would be appropriate to apply to low bioburden drug manufacturing processes.**

Recognizing that the goal of regulatory guidance is protection of patients, low bioburden DS manufacturers must provide layers of protection or barriers to safeguard the process from environmental contamination. This interpretation of the guidance is that not all controls for sterile products (i.e. those referred to in Annex 1) are required for low bioburden biological active substances; the manufacturer should use QRM tools to assess the product process and facility to determine what levels of protection are required and ensure documentation and compliance meet these levels.

Protecting the DS and intermediates through closed processing is a consistent, reproducible and robust approach, more so than attempting to control the environment around open processing. Closing the process yields a lower risk to the manufactured DS, and ultimately to the patient. However, if you are unable to close the process, you need to assess the level of environmental controls your product and process needs. If cleanrooms are deemed appropriate controls, the guidance and principles in Annex 1 should be taken into account.

The Design of Premises/Operations and Grades are key controls in the manufacture of pharmaceuticals; however, it is only one layer of a multi-layer control strategy. Such control strategies should be in alignment with Annex 2 and use QRM principles to ensure that selection of control measures is commensurate with the potential risks to the product and patient. Over-application of Annex 1 guidance to low bioburden DS facilities could result in higher cost and greater operational burdens without any additional benefit of higher quality and safety of product for the patients.



# References

- 1 EudraLex – Volume 4 – *Good Manufacturing Practice (GMP) guidelines* (December 2010)
- 2 EU GMP Annex 2: *Manufacture of Biological Active Substances and Medicinal Products for Human Use* (June 2018)
- 3 EU GMP Annex 1: *Manufacture of Sterile Medicinal Products* (August 2022)
- 4 BioPhorum, *Streamlining environmental monitoring in low bioburden drug substance production: industry interpretation of Annex 1* (2024)
- 5 Bevan, N., et al, *Risk-Based Selection of Environmental Classifications for Biopharmaceutical Operations*, PDA Journal of Pharmaceutical Science and Technology July 2021, 75 (4) 374-390; DOI: <https://doi.org/10.5731/pdajpst.2019.010660>
- 6 ISPE Baseline Guide Volume 6: *Biopharmaceutical Manufacturing Facilities*, 3rd Edition (November 2023)
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- 8 ISO 14644-1:2015 *Cleanrooms and associated controlled environments – Part 1: Classification of air cleanliness by particle concentration*

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